

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour

Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

I A.Sai Swetha (192319071) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph: A



Task: Starting from node A, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration Queue Visited Node

1
2
3
4
5
6
7

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

ANS) 1. Write the BFS traversal order.

The BFS traversal order is:

A → B → C → D → E → F → G

2. Show the queue contents after each iteration.

Iteration	Queue After Iteration
1	[B, C]
2	[C, D, E]
3	[D, E, F, G]
4	[E, F, G]
5	[F, G]
6	[G]
7	[]

BFS Explanation

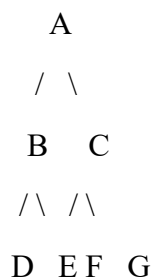
BFS uses a First-In, First-Out (FIFO) queue. Starting with A, its adjacent nodes B and C are inserted into the queue. B is then visited and D and E are added. Next, C is visited and F and G are added. The remaining nodes are visited in the order D, E, F, and G.

Final Answer

BFS Traversal: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$

Queue sequence after each iteration: $[B, C] \rightarrow [C, D, E] \rightarrow [D, E, F, G] \rightarrow [E, F, G] \rightarrow [F, G] \rightarrow [G] \rightarrow []$

2. Depth-First Search (DFS) Using the same graph,



Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from A.

Complete the table.

3. Initial State :

```

1 3 6
5 0 2
4 7 8
  
```

Goal State :

```

1 2 3
4 5 6
7 8 0
  
```

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration Current State Queue Before Expansion Queue After Expansion

1
2
3
4
5

Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

ANS) 1. Breadth-First Search (BFS)

Initial State:

```
1 3 6
5 0 2
4 7 8
```

Goal State:

```
1 2 3
4 5 6
7 8 0
```

BFS explores the state space level by level. The blank-tile move order used here is Up → Down → Left → Right, and repeated states are not inserted into the queue.

BFS Trace – First Five Iterations

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	1 3 6 / 5 0 2 / 4 7 8	[136/502/478]	[106/532/478], [136/572/408], [136/052/478], [136/520/478]
2	1 0 6 / 5 3 2 / 4 7 8	[106/532/478], [136/572/408], [136/052/478], [136/520/478]	[136/572/408], [136/052/478], [136/520/478], [016/532/478], [160/532/478]
3	1 3 6 / 5 7 2 / 4 0 8	[136/572/408], [136/052/478], [136/520/478], [016/532/478], [160/532/478]	[136/052/478], [136/520/478], [016/532/478], [160/532/478], [136/572/048], [136/572/480]
4	1 3 6 / 0 5 2 / 4 7 8	[136/052/478], [136/520/478], [016/532/478], [160/532/478], [136/572/048], [136/572/480]	[136/520/478], [016/532/478], [160/532/478], [136/572/048], [136/572/480], [036/152/478], [136/452/078]
5	1 3 6 / 5 2 0 / 4 7 8	[136/520/478], [016/532/478], [160/532/478], [136/572/048], [136/572/480], [036/152/478], [136/452/078]	[016/532/478], [160/532/478], [136/572/048], [136/572/480], [036/152/478], [136/452/078]

			[130/526/478], [136/528/470]
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Notation: 136/520/478 represents the 3×3 state

1 3 6

5 2 0

4 7 8.

Answers to BFS Questions

1. Which node/state is expanded in each iteration?

Iteration 1: 1 3 6 / 5 0 2 / 4 7 8

Iteration 2: 1 0 6 / 5 3 2 / 4 7 8

Iteration 3: 1 3 6 / 5 7 2 / 4 0 8

Iteration 4: 1 3 6 / 0 5 2 / 4 7 8

Iteration 5: 1 3 6 / 5 2 0 / 4 7 8

2. How many states are generated in total?

Using BFS with duplicate-state checking and the stated move order, 500 unique states are generated/visited by the time the goal is reached. The goal is reached at depth 8. The exact generated-state count can vary if a different move ordering or duplicate-counting convention is used.

3. Complete solution path from initial state to goal

The shortest path contains 8 moves:

Right → Up → Left → Down → Left → Down → Right → Right.

4. Why does BFS always find the shortest solution?

BFS explores all states at depth 1 before depth 2, then depth 3, and so on. Because every legal 8-puzzle move has the same cost, the first time BFS reaches the goal it has used the minimum possible number of moves.

Complete Shortest Solution Path

State 0 – Initial

1 3 6

5 0 2

4 7 8

State 1 – Right

1 3 6

5 2 0

4 7 8

State 2 – Up

1 3 0

5 2 6

4 7 8

State 3 – Left

1 0 3

5 2 6

4 7 8

State 4 – Down

1 2 3

5 0 6

4 7 8

State 5 – Left

1 2 3

0 5 6

4 7 8

State 6 – Down

1 2 3

4 5 6

0 7 8

State 7 – Right

```

1 2 3
4 5 6
7 0 8

```

State 8 – Right (Goal)

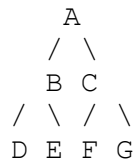
```

1 2 3
4 5 6
7 8 0

```

2. Depth-First Search (DFS) Using the Given Graph

Graph:



Perform DFS starting from A. Assuming left-to-right traversal:

Iteration	Expanded Node
1	A
2	B
3	D
4	E
5	C
6	F
7	G

DFS Traversal: $A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G$ **Conclusion**

BFS is suitable for the 8-puzzle when the objective is to obtain a shortest sequence of moves, because it explores the search space level by level. DFS, on the other hand, follows one branch as deeply as possible before backtracking. For the given graph, the DFS traversal is $A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F \rightarrow G$.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	20	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	30	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Application of Search Algorithm	20	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	20	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	10	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.